

Lesson 16 — Build the Model & If / Then

Matches LEGO Teacher Portal lesson: "Secret Solver".

Objective (I can...): I can build the conditionals model and use an if/then block so it decides what to do.

CS concept: Conditionals (if a condition is true, then act) · **Builds on:** Lesson 15 (events + loops set the stage for decisions).

AI Connection (2 min): AI decides with confidence: "IF I'm 80% sure it's a dog, THEN bark." *First-principles question:* "What should happen if it's only 50% sure?"

Materials: LEGO CS & AI kit + device per pair, trigger object/color card. **This is the unit's build lesson** — build the Conditionals model now and keep it assembled through Lesson 20.

60-minute flow - Warm-up (8): "If... then..." chant — coach says conditions ("if it's raining..."); pairs finish with the action. - **Concept + demo (10):** Demo an if/then: if the sensor sees an object, then light on; otherwise nothing. - **Build/Code (30):** Pairs **build the Conditionals model** (the unit's one build), then program one if/then (object near / color seen → action). *Swap at 15 min.* - **Challenge (7):** Change the condition; test that the action only happens when it's true. - **Share + pack-up (5):** Pairs demo the true case and the "nothing happens" case. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator reads the condition aloud before each run. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — a labeled palette card for the if/then block. *Extension / fast-finisher* — make the "then" a two-step action. *Reteach* — hold the object at/away and narrate "true... false..." while it runs.

Exit check: Pair's model acts only when the condition is true. **At-home (optional):** Write three "if... then..." rules for your house. ~10 min.
